

Sample End Semester Question Paper - 2025

Subject: Signals and Systems | Code: EC301 | Semester: 3

Branch: ECE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. What is the Laplace transform of $u(t)$?
2. What is the Z-transform?
3. State one use of the region of convergence (ROC).
4. Define impulse response.
5. What is a periodic signal?
6. Classify signals into energy signals and power signals.
7. What is the sampling theorem?
8. State one property of a stable LTI system.
9. What is the convolution integral?
10. State one property of the Fourier transform.

PART B - Long questions (8 x 10 = 80 Marks)

1. Explain the Laplace transform. Solve a differential equation using it.
2. Explain the Z-transform and its properties. Find inverse Z-transforms.
3. Explain system analysis using Laplace transform: transfer function, poles and zeros.
4. Compare continuous-time and discrete-time Fourier transforms.
5. Classify signals and systems: energy/power, deterministic/random, LTI with examples.
6. Explain LTI systems. Derive and explain the convolution integral with an example.
7. Find the Fourier series of a square wave and state Parseval's theorem.
8. Explain Fourier transform properties. Find the transform of a given signal.